PRACTICAL - 4

Implementation and Time Analysis of Factorial using Iterative and Recursive Methods

Aim of the Practical:

To implement factorial iteratively and recursively and compare their time and space complexities.

Program:

```
#include <iostream>
using namespace std;
long long factorialIterative(int n) {
    long long fact = 1;
    for (int i = 2; i <= n; i++)
        fact *= i;
    return fact;
}
long long factorialRecursive(int n) {
    if (n <= 1) return 1;
    return n * factorialRecursive(n - 1);
}
int main() {
    int n = 5;
    cout << "Iterative: " << factorialIterative(n) << endl;
    cout << "Recursive: " << factorialRecursive(n) << endl;
}
```

Output:

```
Iterative: 120
Recursive: 120
```

Time Analysis:

Iterative Method - Derivation:

The loop starts at 2 and runs through n.

$$\text{Number of iterations} = n - 1$$

Each iteration performs constant work.

$$T_{iter}(n) = c(n - 1) + c_0 = \Theta(n)$$

Only a fixed number of variables are used, so auxiliary space is $\Theta(1)$.

Recursive method recurrence:

$$T_{rec}(n) = T_{rec}(n - 1) + c$$

Unrolling:

$$\begin{aligned} T_{rec}(n) &= T_{rec}(n - 2) + 2c \\ &= T_{rec}(1) + (n - 1)c \\ &= \Theta(n) \end{aligned}$$

There are n recursive stack frames in the worst case, so auxiliary space is $\Theta(n)$.

Final Time Analysis:

- Iterative Time: $\Theta(n)$, Space: $\Theta(1)$.
- Recursive Time: $\Theta(n)$, Space: $\Theta(n)$.

Result:

Both methods compute factorial correctly. Their time is linear, but recursion uses linear stack space.

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PRACTICAL - 5

Implementation of 0/1 Knapsack using Dynamic Programming

Aim of the Practical:

To solve the 0/1 Knapsack problem using dynamic programming and analyze the DP table construction.

Algorithm:

1. Let dp[i][w] denote the best value using the first i items and capacity w.
2. Initialize row 0 and column 0 with 0.
3. If wt[i-1] > w, set dp[i][w] = dp[i-1][w].
4. Otherwise set dp[i][w] = max(dp[i-1][w], value[i-1] + dp[i-1][w-wt[i-1]]).
5. Return dp[n][W].

Program:

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
int knapsack(int W, const vector<int>& wt, const vector<int>& val) {
    int n = wt.size();
    vector<vector<int>> dp(n + 1, vector<int>(W + 1, 0));
    for (int i = 1; i <= n; i++) {
        for (int w = 0; w <= W; w++) {
            dp[i][w] = dp[i - 1][w];
            if (wt[i - 1] <= w) {
                dp[i][w] = max(dp[i][w],
                               val[i - 1] + dp[i - 1][w - wt[i - 1]]);
            }
        }
    }
    return dp[n][W];
}
int main() {
    vector<int> wt = {1, 3, 4, 5};
    vector<int> val = {1, 4, 5, 7};
    int W = 7;
    cout << knapsack(W, wt, val);
}
```

Output:

```
9
```

Time Analysis:

Derivation of Time Complexity:

Let n be the number of items and W be the knapsack capacity.

The DP table contains (n + 1) rows and (W + 1) columns.

$$\text{Number of states} = (n + 1)(W + 1)$$

The two nested loops visit every non-base state once.

For each state, only constant-time comparisons, additions and max operations are performed.

$$T(n,W) = c \cdot n \cdot (W + 1) + c0$$

$$T(n,W) = \Theta(nW)$$

The table stores (n + 1)(W + 1) integer values.

$$S(n,W) = \Theta(nW)$$

This is pseudo-polynomial because W is a numeric capacity, not the number of bits needed to encode W.

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Final Time Analysis:

- Time Complexity: Θ(nW).
- Space Complexity: Θ(nW).

Result:

The 0/1 Knapsack problem was solved using dynamic programming, and its complexity was derived from the number of DP states.